

JavaScript 1.4

this, Invocation & Object Model

Print-first interview refresher | call site -> receiver -> property -> prototype -> invariant

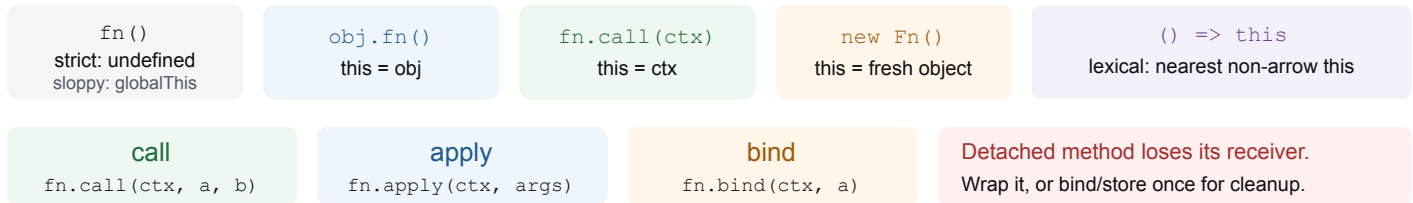
Ask at every access:

Who is the receiver?

Own key, prototype, or trap?

1 | Invocation decides regular this

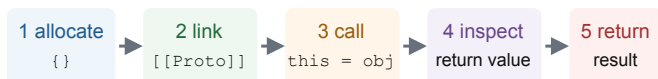
Call site first. Definition site does not bind regular this.



call/apply invoke now; bind returns a function. Bound args come first; later args append.

Re-bind keeps first this; new on a bound target uses a fresh this; call/apply/bind cannot replace arrow this.

2 | new: five-step mental model



Constructor - prototype - instance



Primitive return: ignored. Object/function return: replaces the fresh object.

Arrow: no [[Construct]], no new, no own .prototype.

new.target reveals the constructor-call path.

User.prototype.constructor -> User; inherited/writable, not a type test.

3 | Prototype lookup and writes



Read searches outward. An own key shadows the same inherited key.

Assignment usually creates/updates own data, but an inherited setter may run.

Inherited non-writable data can reject assignment; defineProperty defines own.

Object.getPrototypeOf(x) inspect the internal [[Prototype]]

proto.isPrototypeOf(x) is proto anywhere in x's chain?

x instanceof C is C.prototype in x's chain?

Cross-realm constructors and Symbol.hasInstance can break naive instanceof tests.

Object.setPrototypeOf / __proto__ (legacy): changing a live chain is slow and fragile.

Object.create(null): no Object.prototype.

4 | Property access has independent axes

	key: string symbol	owner: own inherited	enumerable: true false	in: chain Object.hasOwn: own	
API	own strings	own symbols	inherited	enum only	result / use
Object.keys / values / entries	yes	no	no	yes	common own view
getOwnPropertyNames / Symbols	names	symbols	no	no	one key kind each
Reflect.ownKeys	yes	yes	no	no	complete own keys
for...in	yes	no	yes	yes	enumerable strings
spread / Object.assign	yes	yes	no	yes	shallow value copy

5 | Create objects deliberately

Form	instance state	shared behavior / fit
literal	own properties	direct one-off model
factory	own / closure	privacy and composition
constructor	assigned to this	Ctor.prototype methods
Object.create	initialize explicitly	chosen delegate object
class	fields on instance	methods on Class.prototype

Class is prototype syntax. Prefer composition when an is-a chain adds friction.

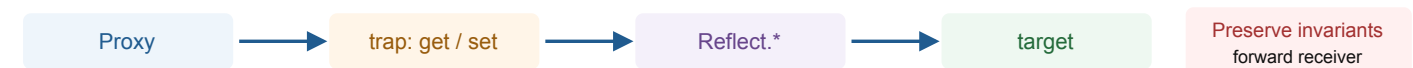
6 | Descriptors and integrity levels

data descriptor		accessor descriptor	
value writable enumerable configurable		get set enumerable configurable	
level	add	delete/configure	write data
normal	yes	yes	yes
preventExtensions	no	yes	yes
seal	no	no	if writable
freeze	no	no	no*

defineProperty omitted flags default false. Rejected strict writes throw.

* shallow: nested objects stay mutable; accessor setters may still act.

7 | Proxy intercepts object operations; Reflect forwards their default semantics



Use for validation, virtualization or membranes when interception is required; keep traps minimal and forward with Reflect.